

Cooking Fuel Data Tool — User Guide

This guide walks through every screen and control in the Cooking Fuel Data Tool. It is intended for analysts and modellers who will use the tool.

The tool always opens to Outputs with the default selections applied. The two top-level views (Outputs / Inputs) sit as folder tabs at the top of the body; the sidebar on the left holds the filters and applies to both views.

Select data parameters
Pick countries, fuels, areas, and a year range
— then view the relevant data on the right.

Filters
Countries
Choose options
Fuel types
charcoal x coal x
electric x fuelwood x
gas x kerosene x
Areas
urban x rural x
overall x
End year
2050

Cooking Fuel Data Tool

Generate tables projecting country-specific cooking fuel demand between 1990 and 2050

Select at least one country in the sidebar to view data.

Default View: first select at least one country to see data.

Section 1 — Default Output: Total Fuel Consumption

On launch you see the Total fuel consumption sub-tab inside the Outputs view. The big table is the headline output of the tool: estimated fuel users and total consumption for every selected (country, area, fuel, year) combination.

Select data parameters

Pick countries, fuels, areas, and a year range — then view the relevant data on the right.

Filters

Countries

Argentina

Fuel types

charcoal coal electric fuelwood gas kerosene

Areas

urban rural overall

End year

2050

Cooking Fuel Data Tool

Generate tables projecting country-specific cooking fuel demand between 1990 and 2050

See / edit input data

Outputs

Total fuel consumption Total emissions

Total fuel consumption

Total rows: 918

iso3	country	area	fuel	year	num_fuel_users_thousands	per_capita_fuel_cons	fuel_cons_tons
ARG	Argentina	overall	charcoal	2000	0	0.16	0
ARG	Argentina	overall	charcoal	2001	0	0.16	0
ARG	Argentina	overall	charcoal	2002	0	0.16	0
ARG	Argentina	overall	charcoal	2003	0	0.16	0
ARG	Argentina	overall	charcoal	2004	0	0.16	0
ARG	Argentina	overall	charcoal	2005	0	0.16	0
ARG	Argentina	overall	charcoal	2006	0	0.16	0
ARG	Argentina	overall	charcoal	2007	0	0.16	0
ARG	Argentina	overall	charcoal	2008	0	0.16	0
ARG	Argentina	overall	charcoal	2009	0	0.16	0
ARG	Argentina	overall	charcoal	2010	0	0.16	0
ARG	Argentina	overall	charcoal	2011	0	0.16	0
ARG	Argentina	overall	charcoal	2012	0	0.16	0

Showing first 500 of 918 rows. Units note: num_fuel_users_thousands is in thousands of people (UN population convention — multiply by 1,000 for absolute people). fuel_cons_tons is in absolute tons (the calculation applies a ×1,000 rescale internally). per_capita_fuel_cons units vary by fuel — MWh/person-year for electric, oven-dry tons/person-year for fuelwood and imp_fuelwood, and tons/person-year for the remaining fuels. fuel_cons_tons inherits these per-fuel units. Charcoal rows: fuel_cons_tons is reported in tons of fuelwood-equivalent biomass (charcoal mass × 6 kiln-yield factor — set in the Per-capita rates input tab), not charcoal at the stove.

Fuel-share data: Updated UN WHO data from O. Stoner (methods: Stoner et al. 2021)

Per-capita data: DFloess, Grieshop, Puzzolo, et al. (2023)

Filename (without extension)

cooking_fuel_output_2000_2050

Download Excel Workbook

Download CSV

Cooking Fuel Data Tool - Projecting cooking fuel demand from 1990 to 2050

Default view once a country is or countries are selected.

1.1 Sidebar filters

The left-side panel controls what data the rest of the app shows. Changes apply immediately on rerun — no "apply" button needed.

Control	What it does	Notes
Countries	Multi-select of all countries present in the fuel-share dataset	Required — the body shows a prompt to pick at least one country before any data is rendered.

Fuel types	Multi-select of fuel categories: fuelwood, charcoal, coal, gas, kerosene, electric, biogas, ethanol, pellets, imp_fuelwood, imp_charcoal	Defaults to the six fuels covered by the UN/WHO household survey (fuelwood, charcoal, coal, gas, kerosene, electric). The other five — biogas, ethanol, pellets, imp_fuelwood, imp_charcoal — are not in that survey, so they carry a 0% population share everywhere and start deselected. Select one here and enter shares in the Fuel shares tab (§ 2.1) to bring it into the output.
Areas	Multi-select: urban, rural, overall	overall is derived as urban + rural (see § 1.3). Default is all three.
End year	Slider snapping to one of 2020 / 2030 / 2035 / 2040 / 2050	Start year is fixed at 2000. Default end year is 2050.

1.2 The output table

The table is a single row per (iso3, country, area, fuel, year) combination. Columns:

Column	Meaning	Units
iso3, country, area, fuel, year	Identifiers — area is one of urban/rural/overall; fuel is lowercase	—
num_fuel_users_thousands	Number of people using this fuel in the given (country, area, year)	Thousands of people (UN WPP convention)
per_capita_fuel_cons	Per-capita consumption rate from the per-capita dataset	Varies by fuel: MWh/person-year for electric; oven-dry tons/person-year for fuelwood/imp_fuelwood; tons/person-year for others
fuel_cons_tons	Total consumption: num_fuel_users_thousands × per_capita_fuel_cons × 1,000. Charcoal rows are additionally multiplied by the kiln yield (§ 3.2), and wood/charcoal rows by the non-residential uplift if it is enabled (§ 3.3) — so for	Absolute tons (rounded to whole numbers in the display and exports)

	those fuels the plain three-term identity does not hold.	
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The region column exists in the backend (used to join per-capita rates) but is hidden from the display so the table stays readable.

Units quick-reference

num_fuel_users_thousands is in thousands of people, but fuel_cons_tons is in absolute tons. The calculation applies a $\times 1,000$ rescale internally so the consumption value lands in absolute tons (more useful for emissions and policy). Because of that, the identity $\text{users} \times \text{per_capita} = \text{fuel_cons_tons}$ does NOT hold without the additional $\times 1,000$ — see the inline caption beneath the table and the metadata sheet of the Excel export for the explicit note.

1.3 How overall is computed

The overall row for each (country, fuel, year) is computed as the sum of the urban and rural rows for that combination — not taken from the source's "Overall" series. This guarantees $\text{overall.num_fuel_users_thousands} = \text{urban.num_fuel_users_thousands} + \text{rural.num_fuel_users_thousands}$ and the same for fuel_cons_tons. The source's Overall rows from the WHO/Stoner dataset are intentionally discarded; the modelled "Overall" share doesn't always equal the population-weighted average of the urban and rural shares, and the tool prefers the arithmetic identity for consistency.

1.4 Downloading the data

Below the table are two download buttons side-by-side:

Button	Format	Contents
Download Excel Workbook	.xlsx	Two sheets: Metadata & Sources (run parameters, citations, units notes) and Fuel Consumption Data (the table).
Download CSV	.csv	Single file. The first two rows are a parameter preamble — row 1 variable names, row 2 the matching values — carrying the run parameters and total demand per fuel (see "The CSV parameter preamble" below). Row 3 onward is the data table (header + rows). Designed for ingestion by downstream tools that expect a named-parameter header above the

		data.
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The text input above the buttons sets the filename stem — both downloads use it (with .xlsx and .csv extensions respectively). Illegal filename characters are stripped automatically.

The CSV parameter preamble

The CSV export opens with a two-row parameter preamble: row 1 holds variable names, row 2 holds the matching values, column for column. The consumption data table begins on row 3. This lets a downstream tool read the run's settings and headline results straight off the top of the file without re-aggregating the data rows.

end_year,efchratio,nonres_wood_pct,nonres_charcoal_pct,demand_yr_start,demand_yr_end,demand_fw,demand_ch,demand_coal,demand_gas,demand_kero,demand_elec,demand_biogas,demand_eth,demand_pel,demand_impfw,demand_impch

2050,6,10,20,2026,2035,158876572,7045763,0,8530677,942834,148107,0,0,0,0,0

The parameter columns:

Column	Meaning	Units
end_year	Last projection year, from the sidebar slider.	year
efchratio	Wood-to-charcoal kiln yield (§ 3.2), already applied to charcoal rows.	kg wood / kg charcoal
nonres_wood_pct	Non-residential uplift applied to fuelwood and imp_fuelwood. 0 when the option is switched off — a 0% uplift is equivalent to not applying it, so there is no separate on/off flag.	percent
nonres_charcoal_pct	Non-residential uplift applied to charcoal and imp_charcoal. 0 when switched off.	percent
demand_yr_start, demand_yr_end	The year window the demand_* totals were summed over. Normally 2026–2035; clipped if the end-year filter stops earlier.	year
demand_<fuel>	Total consumption of that fuel over the demand window — one value per fuel. See the mapping table below.	tons (MWh for demand_elec)

Demand columns

Each demand_<fuel> value is the total consumption of that fuel summed over the demand window, across the countries currently selected in the sidebar, using the overall area rows (overall = urban + rural, so this is a national total and is not double-counted against the urban and rural rows). Values reflect everything applied to the data rows below: inline edits, the kiln yield, and the non-residential uplift.

All eleven fuels are always written, in the fixed order below, with 0 for any fuel that is deselected in the sidebar or carries a zero share. The preamble layout is therefore identical between exports, so a downstream parser can rely on a constant schema.

Column	Fuel	Units
demand_fw	fuelwood	oven-dry tons
demand_ch	charcoal	tons of fuelwood-equivalent biomass (kiln yield applied)
demand_coal	coal	tons
demand_gas	gas	tons
demand_kero	kerosene	tons
demand_elec	electric	MWh
demand_biogas	biogas	tons
demand_eth	ethanol	tons
demand_pel	pellets	tons
demand_impfw	imp_fuelwood	oven-dry tons
demand_impch	imp_charcoal	tons of fuelwood-equivalent biomass

Note: the 2026–2035 window is fixed in the tool, not derived from today's date. If the end-year slider is set below 2035 the window is truncated (for example 2026–2030 at an end year of 2030); the app shows a warning and the actual span is always recorded in demand_yr_start / demand_yr_end, so read those rather than assuming a full decade.

About the kiln yield in the CSV preamble

The efchratio value in the CSV preamble is the wood-to-charcoal kiln yield set in the Per-capita rates input tab (§ 3). It applies to charcoal rows of fuel_cons_tons: those rows are multiplied by the kiln yield so they represent the upstream wood biomass burned to produce the charcoal — the right unit for forest-impact / biomass-supply analyses, not stove-side charcoal mass.

Section 2 — Inputs: Fuel Shares

Click "See / edit input data" at the top to switch from outputs to the inputs view. The Inputs view has two sub-tabs; Fuel shares is the first.

Cooking Fuel Data Tool

Generate tables projecting country-specific cooking fuel demand between 1990 and 2050

 See / edit input data

 Outputs

Fuel shares Per-capita rates

Filtered fuel-share data

% of population using each fuel, by country, area, fuel, and year. Edit cells directly or paste from Excel, then click Save.

Total rows: 612

iso3	country	area	fuel	year	 population_share
ARG	Argentina	urban	kerosene	2000	0.0000
ARG	Argentina	urban	kerosene	2001	0.0000
ARG	Argentina	urban	kerosene	2002	0.0000
ARG	Argentina	urban	kerosene	2003	0.0000
ARG	Argentina	urban	kerosene	2004	0.0000
ARG	Argentina	urban	kerosene	2005	0.0000
ARG	Argentina	urban	kerosene	2006	0.0000
ARG	Argentina	urban	kerosene	2007	0.0000
ARG	Argentina	urban	kerosene	2008	0.0000
ARG	Argentina	urban	kerosene	2009	0.0000

 Save edits

 Data Source: Updated UN WHO data from O. Stoner (methods: Stoner et al. 2021)

>  View Full Citations

Customize the proportional fuel-use data

 Upload custom dataset

 Customize year projections

Cooking Fuel Data Tool - Projecting cooking fuel demand from 1990 to 2050

Select/see input data tab → Fuel shares

2.1 The editable share table

The table shows the fraction of population using each fuel for every (iso3, country, area, fuel, year) in the current filter selection. The editable column is population_share (values 0–1); all identifier columns are locked.

Five fuels — biogas, ethanol, pellets, imp_fuelwood and imp_charcoal — are absent from the UN/WHO household survey that supplies the default shares. They are injected with a population share of 0 for every country, area and year so they can be edited here like any other fuel: select one in the sidebar, enter shares in this table, and it flows through consumption and emissions normally. Per-capita rates and emission factors for them are already present in the reference data, so nothing else needs supplying.

To edit:

1. Click a cell in the population_share column and type a new value, OR paste a column from Excel.
2. Click Save edits to commit. The status banner confirms the row count applied.

Edits are persistent for the rest of the session: they flow into the consumption calculation the moment you save, and the metadata sheet of any subsequent Excel export records that in-app edits were made.

Note: the editor switches to a read-only preview if the current selection produces more than 8,000 rows. Narrow your sidebar filters (fewer countries or a shorter year range) to drop below that threshold and re-enable editing.

2.2 Data source ribbon and citations

Below the table is a coloured ribbon showing the current active data source — blue for the default Stoner et al. (2021) data, yellow when a custom dataset has been loaded or in-app edits have been applied. Expand "View Full Citations" to see the bibliographic entries for whatever is currently in use, including any custom-projection citation you provided.

2.3 The four action buttons

Below the citations are four buttons; the last two only appear when the corresponding state is active.

Button	What it does
Upload custom dataset	Opens a modal to upload your own fuel-share CSV/XLSX (replaces the default Stoner et al. data wholesale). The required columns are iso3, country, region, area, fuel, year, population_share. Region is re-derived from iso3 on upload so user-supplied region values

	are normalised.
Customize year projections	Opens a modal to set custom values for a specific year (e.g. 2030); the rest of the timeseries is linearly interpolated between the 1990 baseline and the custom year, then to 2050. Provide either the values inline in a data editor or by uploading a CSV. Either way, a data-source citation is required.
Revert to default WHO data	Only visible after a custom dataset has been loaded. Restores the bundled Stoner et al. data.
Clear custom projections	Only visible after custom-year projections are active. Removes them and returns to the baseline projections.

Note on the 0-share fuels: interpolation anchors to the first and last baseline years, so setting a custom value for a fuel whose baseline is 0 everywhere produces a ramp up to your custom year and then a decay back to 0 by 2050. To hold such a fuel at a steady share, edit the values directly in the Fuel shares table (§ 2.1) instead.

Custom upload — minimum requirements

- File must be .csv or .xlsx with the header row matching iso3, country, region, area, fuel, year, population_share (case-insensitive; the loader lowercases column names).
- population_share values must be between 0 and 1.
- A free-text data-source description is required at upload time (it shows up in the citations expander and the Metadata sheet of every subsequent Excel export).
- Any Overall rows in the upload are dropped; overall is always derived as urban + rural downstream. Use lowercase urban/rural (case is normalised automatically).
- Fuels your file omits are backfilled automatically: any of biogas, ethanol, pellets, imp_fuelwood or imp_charcoal that is missing is added at a 0 share, so the fuel list stays consistent with the default dataset.

Section 3 — Inputs: Per-Capita Rates

The second Inputs sub-tab.

Cooking Fuel Data Tool

Generate tables projecting country-specific cooking fuel demand between 1990 and 2050

 See / edit input data

 Outputs

Fuel shares Per-capita rates

Per-capita rates by region and fuel

Each value is per fuel user per year. Rows from the source data with no region apply globally — they have been expanded to all regions. Edit cells directly or paste from Excel, then click Save.

Total rows: 30

fuel	region	pc_fuel	pc_fuel_units
charcoal	east_asia	0.1800	tons/person-year
charcoal	latam	0.1600	tons/person-year
charcoal	other	0.1600	tons/person-year
charcoal	south_asia	0.1300	tons/person-year
coal	south_asia	0.2400	tons/person-year
coal	ssa	0.2400	tons/person-year

 Save edits

Wood-to-charcoal kiln yield

6

Why this matters: charcoal kilns waste most of their input wood as heat. A factor of 6 means producing 1 ton of charcoal consumes ~6 tons of dry wood. In the Outputs tab, charcoal rows of `fuel_cons_tons` count this upstream wood biomass, not charcoal mass at the stove. Wood-to-charcoal conversion efficiencies vary widely. The recent [UNFCCC INRB assessment](#), which was used to derive the data in UNFCCC's [Tool33](#) used 6:1, which we use here as a default. Any alternate entry should be supported by a well-documented field-based assessment.

 Per Capita Data Source: DFloess, Grieshop, Puzzolo, et al. (2023)

>  View Per Capita Citation

Customize the per-capita rates

 Upload custom rates

Select/see input data tab → Per-capita rates

3.1 The editable rate table

Each row gives the per-capita annual fuel consumption for one (region, fuel) pair. The editable column is pc_fuel; fuel, region, and pc_fuel_units are locked. The table is filtered to the fuels selected in the sidebar so you only see the rows relevant to your current run.

Per-capita rates are keyed on (region, fuel) — not on area or year — so a single edit applies to every (country, area, year) row in that region/fuel combo. This is by design: per-capita rates are intended as regional structural assumptions, not country-level observations.

The pc_fuel_units column shows the unit each rate is expressed in (varies by fuel — MWh/person-year for electric, oven-dry tons/person-year for fuelwood/imp_fuelwood, tons/person-year for the rest).

Click Save edits to apply your changes. As with the share editor, edits are persisted for the rest of the session and noted in the Excel metadata sheet.

3.2 Wood-to-charcoal kiln yield (efchratio)

Below the per-capita table is the Wood-to-charcoal kiln yield input — a single numeric field. This is the value exported as efchratio in the CSV preamble (§ 1.4).

Detail	Value
What it represents	Kilograms of dry wood needed to produce 1 kg of charcoal in a kiln.
Where it's applied	The fuel_cons_tons column for charcoal rows only. After computing the stove-side charcoal mass, the value is multiplied by the kiln yield to give the upstream wood biomass.
Default	6 (typical traditional earthen-kiln yield in sub-Saharan Africa — FAO regional reference).
Typical range	3–5 for more efficient kilns; 1 if you want to keep charcoal rows as stove-side mass.

Why this matters: charcoal kilns waste most of their input wood as heat. A factor of 6 means that producing 1 ton of charcoal consumes ~6 tons of dry wood — the right number for forest-impact / fNRB analyses. The caption block under the input links to the UNFCCC fNRB assessment that informed the default.

3.3 Non-residential wood & charcoal consumption

Below the kiln yield is an optional adjustment for non-residential demand. The fuel-share data covers household cooking only; this uplift accounts for restaurants, schools, prisons, bakeries and other institutional or small-commercial users drawing on the same wood and charcoal supply.

Detail	Value
How to enable	Tick "Include non-residential wood and charcoal consumption". Unticked by default, so outputs are household-only unless you opt in.
Controls	Two percentage inputs appear when ticked: wood uplift (default 10%) and charcoal uplift (default 20%).
Where it's applied	fuel_cons_tons is multiplied by $(1 + \text{uplift}/100)$ — the wood figure for fuelwood and imp_fuelwood, the charcoal figure for charcoal and imp_charcoal. Applied after the kiln yield, so charcoal scales on its fuelwood-equivalent basis.
Effect on other columns	num_fuel_users_thousands is unchanged — this is extra demand on the same fuel supply, not extra people. Emissions follow the uplifted consumption automatically.
Where it's recorded	A banner above the consumption table while active; the nonres_wood_pct and nonres_charcoal_pct columns of the CSV preamble; and the "Non-Residential Consumption" row of both Excel metadata sheets.

The defaults of 10% for wood and 20% for charcoal are starting points, not measured values — the non-residential share varies widely by country and settlement type. Any alternative entry should be supported by a documented field-based assessment.

3.4 Custom per-capita upload

If the default per-capita data is missing or you want to replace it, the Upload custom rates button (visible at the top of the tab when no default is loaded, and via a modal action when it is) accepts a .csv or .xlsx with columns fuel, region, pc_fuel, pc_fuel_units. Rows with an empty region value are treated as global and expanded into every region at load time. As with the share upload, a data-source citation is required and recorded in the metadata sheet.

Section 4 — Emissions Output

Back in the Outputs view, the second sub-tab is Total emissions. The emissions calculation uses the consumption results from Section 1 as input: $\text{total_GHG} = \text{fuel_cons_tons} \times \text{em_intens_GHG}$ for each of CO₂, CH₄, and N₂O.

Cooking Fuel Data Tool

Generate tables projecting country-specific cooking fuel demand between 1990 and 2050

 See / edit input data

 Outputs

Total fuel consumption **Total emissions**

Total emissions

Table view

☒ Per-fuel detail ☐ Country / area summary (CO₂-eq)

Total rows: 918

iso3	country	area	fuel	year	fuel_cons_tons	em_intens_CO2	em_intens_CH4	em_intens_N2O	total_CO2	total_CH4	total_N2O
ARG	Argentina	overall	charcoal	2000	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2001	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2002	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2009	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2010	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2011	0	6.2081	2.5684	0.1065	0	0	0
ARG	Argentina	overall	charcoal	2012	0	6.2081	2.5684	0.1065	0	0	0

Showing first 500 of 918 rows. Calculation: $\text{total_GHG} = \text{fuel_cons_tons} \times \text{em_intens_GHG}$ per row. Units: all **total_*** columns are in absolute tons of GHG. Non-electric: intensities are mass ratios (kg GHG / kg fuel = tons/ton). Electric: source values are gCO₂/kWh and divided by 1000 at load (= tons CO₂ / MWh), so the multiplication directly yields tons. CH₄ and N₂O for electricity are 0 — country-level data not available; those gases contribute <5% of CO₂-eq for most grids.

 **Non-electric emissions intensity:** per-fuel intensities for all countries (Electricity row ignored — handled separately)
 **Electricity emissions intensity:** per-country Combined Margin grid emission factor (CO₂ only; CH₄ and N₂O set to 0). Source: [UNFCCC — IFI TWG List of Methodologies](#).

Filename (without extension) 

cooking_fuel_emissions_2000_2050

 Download Emissions Excel Workbook

Output data tab → Emissions from fuel use

4.1 Two views — toggle at the top

A radio control at the top of the tab switches between two views of the same emissions calculation:

View	Row granularity	Columns
Per-fuel detail	(iso3, country, area, fuel, year)	fuel_cons_tons, em_intens_CO2/CH4/N2O,

		total_CO2/CH4/N2O
Country / area summary (CO2-eq)	(iso3, country, area, year) — summed across all fuels	total_CO2, total_CH4, total_N2O, total_CO2eq

The summary view computes:

$$\text{total_CO2eq} = \text{total_CO2} + 28 \times \text{total_CH4} + 265 \times \text{total_N2O}$$

using IPCC AR5 GWP-100 factors — the UNFCCC Enhanced Transparency Framework default for reporting after 2024.

4.2 Units and electricity treatment

- All total_* columns are in absolute tons of GHG.
- For non-electric fuels, intensities are mass ratios (kg GHG / kg fuel = tons/ton) sourced from the per-fuel intensity file.
- For electricity, per-country Combined Margin grid emission factors (from the UNFCCC IFI TWG methodology list) are used; the loader divides the source gCO2/kWh by 1,000 so the in-memory value lines up dimensionally with fuel_cons_tons for electric rows (which is in MWh).
- CH4 and N2O for electricity are set to 0 because country-level data isn't available; for most grids these gases contribute less than 5% of CO2-equivalent, so the omission has limited impact on the summary view but is an important caveat to keep in mind for grids where biomass cofiring is significant.

The on-screen caption below each table restates these units and the calculation explicitly.

4.3 Missing-intensity diagnostic

If any rows in the per-fuel detail have no matching emissions intensity (likely cause: a fuel name in your share/per-capita data doesn't match anything in the intensity files), a yellow "N rows have no matching emissions intensity" expander appears under the table. Open it to see the count grouped by fuel and identify which fuel name needs reconciling.

4.4 Downloading the emissions data

The Download Excel Workbook button at the bottom of the emissions tab produces a three-sheet workbook:

Sheet	Content
Metadata & Sources	Run parameters, intensity sources, GWP factors, units note.
Total Emissions	The per-fuel detail table (regardless of which view is on screen).
Summary by Country-Area	The aggregated CO2-eq summary.

The filename stem is independent from the consumption export — it starts as `cooking_fuel_emissions_<start_year>_<end_year>` but is editable in the text input above the button.

There is no CSV bundle for the emissions tab at present; the Excel workbook is the only download option here.

Appendix — Quick reference

Where each parameter lives

Parameter	Where to set it	Where it shows up in exports
selected_countries, selected_fuels, selected_areas	Sidebar	Metadata sheet, CSV preamble (in CSV bundles), filter applied to data rows
start_year	Fixed at 2000 — not user-editable	Metadata sheet "Year Range"
end_year	Sidebar slider	Metadata sheet "Year Range", end_year in CSV preamble; also bounds the demand window
efchratio (kiln yield)	Per-capita rates tab → Wood-to-charcoal kiln yield	Metadata sheet, efchratio in CSV preamble, applied to charcoal rows of fuel_cons_tons
Non-residential uplift (wood %, charcoal %)	Per-capita rates tab → Non-residential wood & charcoal consumption	Metadata sheet "Non-Residential Consumption"; nonres_wood_pct and nonres_charcoal_pct in CSV preamble (0 when off)
Decadal demand window (2026–2035)	Fixed in the tool; truncated by the end-year slider	Metadata sheet "Decadal Demand Window"; demand_yr_start / demand_yr_end in CSV preamble
Total demand per fuel	Computed — not user-set	demand_fw, demand_ch, ... (all 11 fuels) in CSV preamble
Custom share dataset / projections	Fuel shares tab → action buttons	Metadata sheet records the source description and citation
Per-capita inline edits	Per-capita rates tab → data editor + Save	Metadata sheet flag "Per Capita — In-App Edits: Yes"
Share inline edits	Fuel shares tab → data editor + Save	Metadata sheet flag "Population Share — In-App Edits: Yes"

Conventional units cheat sheet

Quantity	Unit
num_fuel_users_thousands	thousands of people
per_capita_fuel_cons	varies by fuel (see column pc_fuel_units on the per-capita tab)
fuel_cons_tons	absolute tons
total_CO2, total_CH4, total_N2O, total_CO2eq	absolute tons of GHG
efchratio	kg wood / kg charcoal (dimensionless)
population_share	fraction in [0, 1]
demand_<fuel> (CSV preamble)	tons; MWh for demand_elec; fuelwood-equivalent tons for charcoal
nonres_wood_pct, nonres_charcoal_pct	percent (0 when the uplift is off)